Davis Germain

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Education

Carnegie Mellon University - School of Computer Science **Pittsburgh, PA**

B.S. - Human-Computer Interaction with Concentration in Computer Graphics Expected May 2028

GPA: 3.55

Relevant Coursework*:* Software Structures for User Interfaces, Web Application Development, Principles of Imperative Computation, User-Centered Research and Evaluation, Design for Artificial Intelligence Products, Principles of Imperative Computation, Introduction to Computer Systems, Interaction Design Fundamentals

Technical Skills

Languages: C, Python, Java, JavaScript, HTML, CSS, SML, C++, React

Concepts: Object-Oriented Programming (OOP), Web Development, UI/UX, Human-centered Design, Rapid Prototyping, AGILE Workflow, Computer Graphics, AI-Assisted Design

Tools: Git, Linux, Unity, Miro, Figma, AWS (EC2), Django

Experience

Software Engineering Intern – Quicksilver Labs **May 2026 - August 2026**

- Conducted research into Time-Series Foundation Models (TSFMs) to investigate their use cases and applicability to company work (TSFM Research)
- Designed and built a schema-alignment review tool replacing a manual spreadsheet workflow, targeting a 25% cut in review time (Data UI Project)
- Built and deployed a platform that ingests newsletters, scores them, and exports a reviewed digest to Word to replace a manual flow (Newsletter Project)
- Made VC deal intake document-aware using embedding-similarity retrieval over S3-stored document chunks, injecting relevant context into LLM prompts to deliver to user (VC OS Project)

Research Assistant - Human Computer Interaction Institute **August 2026**

- Helped develop and test a set of design principles for human-agent interaction and how well current agent systems follow them in practice

Projects

Scotty3D: A 3D Modeling Software **August-December 2026**

A 3D modeling software similar to Blender where users can create models for renders and other uses

- Implemented a CPU rasterizer capable of various interpolation algorithms, mip-mapping, and supersampling
- Created a series of local and global operations on halfedge meshes, such as extrude face or Catmull-Clark subdivision

Adaptive UI Redesign - Canvas QuickClick **February 2026**

A redesign of the company "Canvas" UI to adapt to each user with simple AI

- Uses a linear regression model and front-end javascript changes to help users navigate to their most prioritized classes, resulting in an estimated 10-15x ROI

Dynamic Memory Allocator (malloc) **October 2025**

Key programming tool used in C to manage computer memory and optimize heap usage

- Implemented a dynamic memory allocator in C using segregated free lists, applying data structures and algorithms to manage heap memory efficiently
- Tested and debugged allocator using custom trace files and performance benchmarks, achieving 68% heap memory utilization and 14,000 KOPS throughput

Activities

ScottyLabs Events Committee Member **May 2026 - Present**

- Planned logistics and set judges for each track in the ScottyLabs NOVA Hackathon for 400+ participants